CIS 635 Knowledge Discovery & Data Mining

ML Models: Ensemble Learning

Classification Models

- *k-NN*
- *Decision Tree*
- *Logistic Regression*
- **Random Forest Classifier**
- **Boosting Classifiers**
- Support Vector Machines (SVMs)
- Naive Bayes

Ensemble Learning/Meta Learning

Bias-Variance Tradeoff

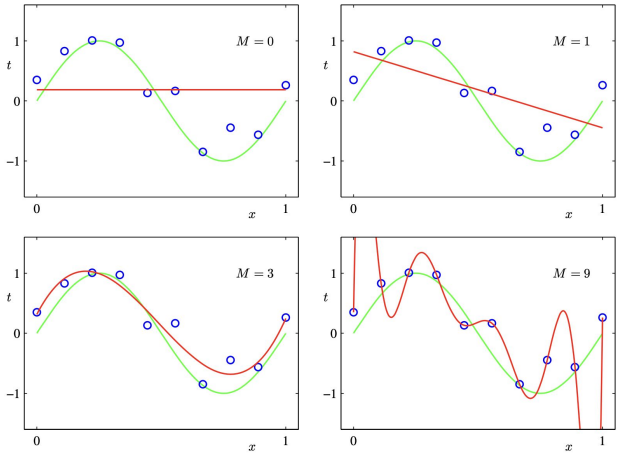
Machine learning models must balance:

- **Bias** → Error due to overly simple assumptions
    - High bias → Underfitting
- **Variance** → Error due to sensitivity to training data
    - High variance → Overfitting

Bias

Bias (Underfitting)

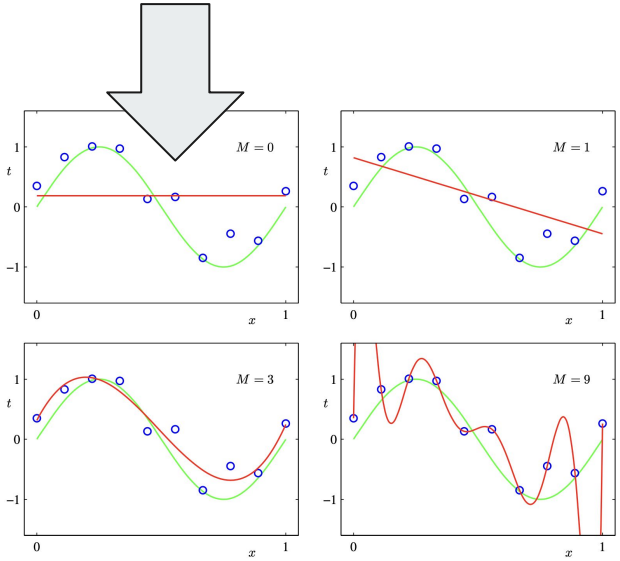
- Model makes **strong assumptions** about the data → misses patterns
- **Example:**
  - True function: $y = x^2$
  - Model: Linear regression ($y = ax + b$)
  - Prediction: Straight line → cannot capture curve → high bias



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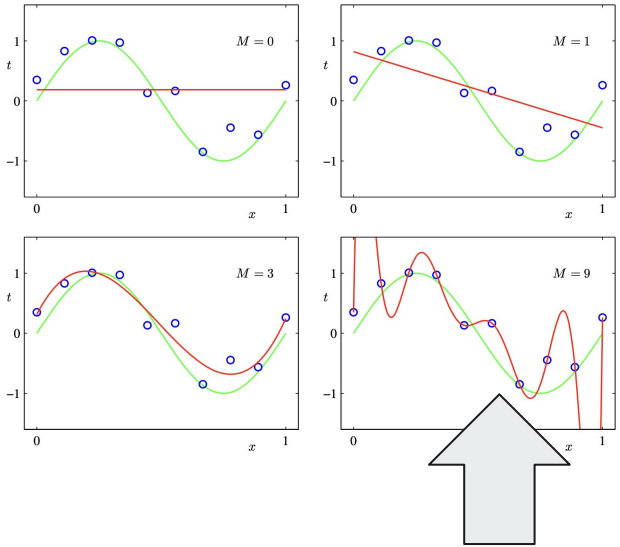
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Variance

Variance (Overfitting)

- Model is **too sensitive** to training data → learns noise
- **Example:**
  - True function: $y = x^2$
  - Model: 10th-degree polynomial
  - Prediction: Fits all training points exactly → wild fluctuations → high variance



Bias-Variance Tradeoff

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Ensemble Learning/Meta Learning

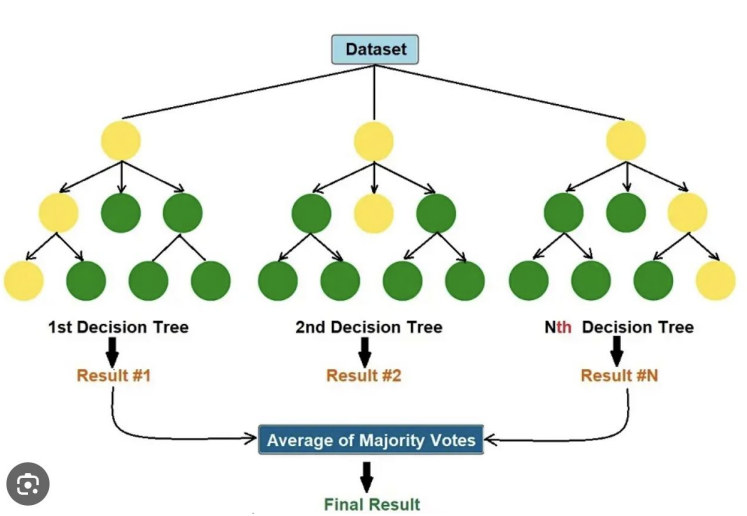
- Ensemble (more than one model)
  - **Bagging**
    - Averaging (RF)
    - Primarily Reduces Variance
  - **Boosting**
    - Gradual improvement over weak learners (Adaboost/XGBoost)
    - Primarily Reduces Bias

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Random Forest

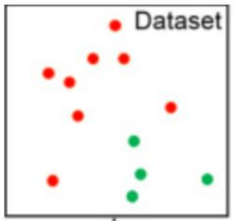
- Instead of one, we have many (but finite) Trees



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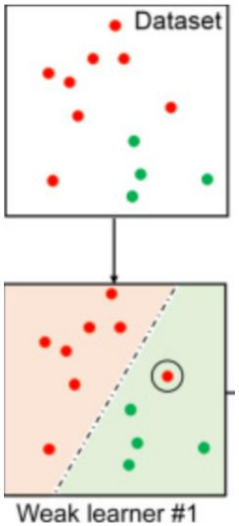
Boosting

- Combination of some weak models
- Power of many



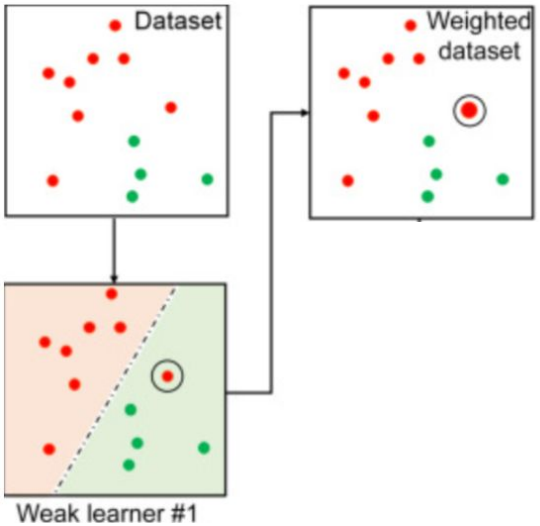
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Boosting



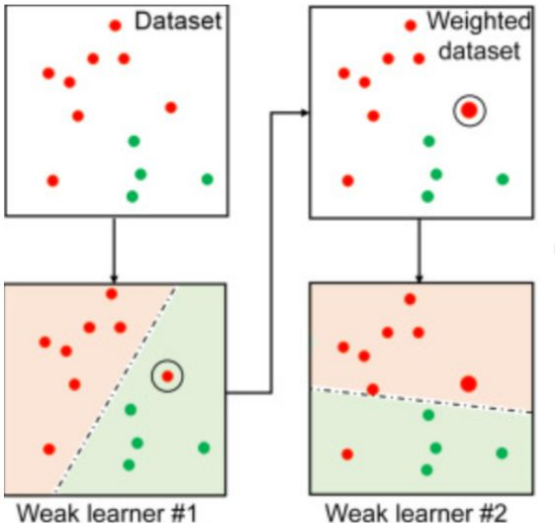
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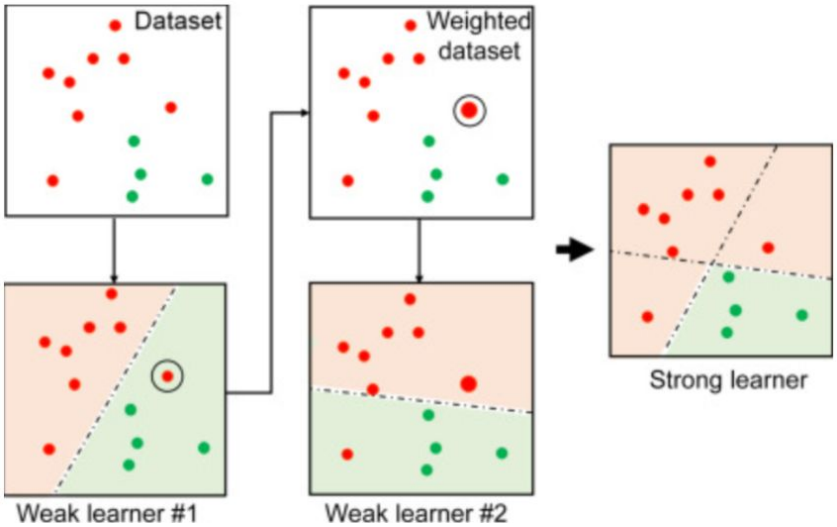
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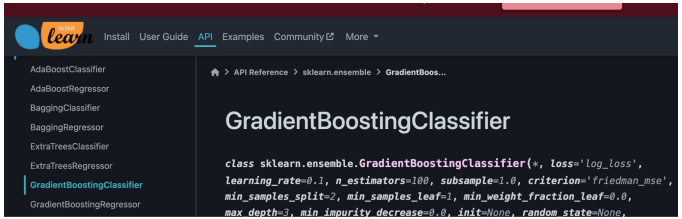


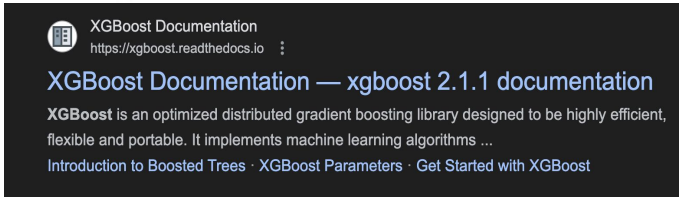
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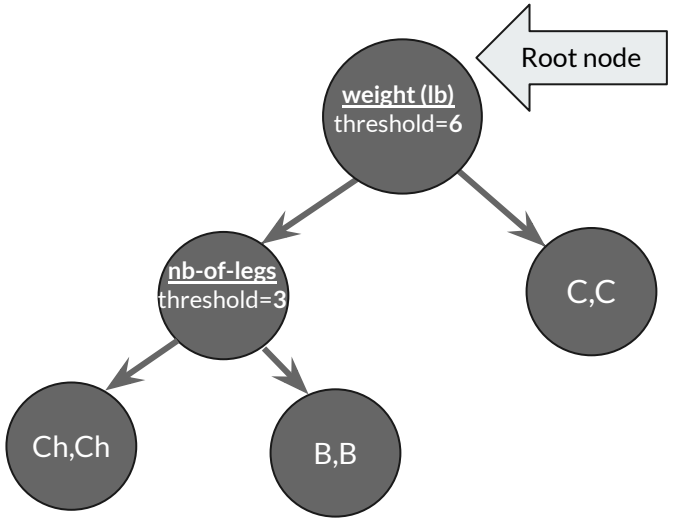


Notebook extension/presentation

- See notebooks section (Blackboard)

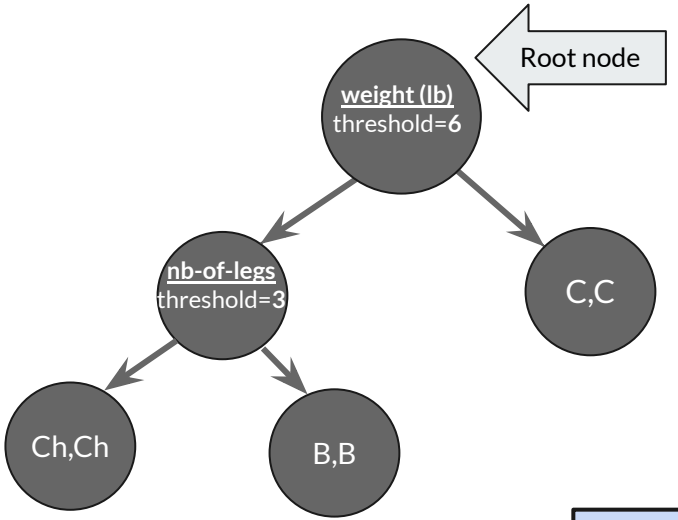
QA

Decision Tree



nb of legs	weight (lb)	animal
4	2.1	Bunny
4	7	Cat
4	1.7	Bunny
4	9	Cat
4	2.75	Bunny
2	2.5	Chicken
2	3	Chicken

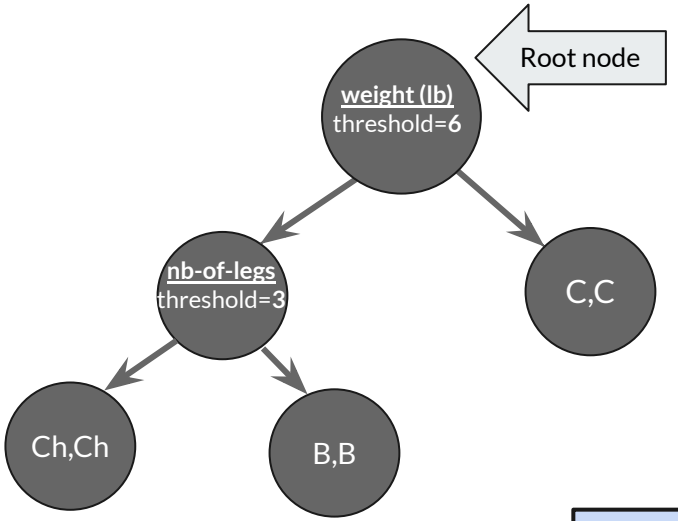
Decision Tree



Test case

2	4	??

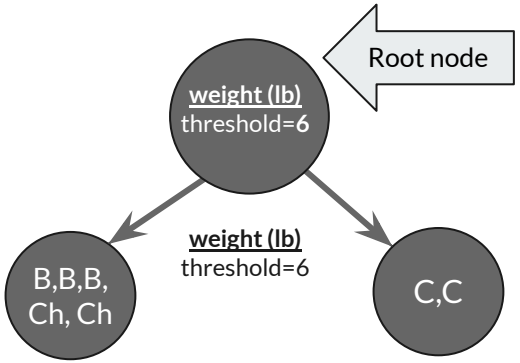
Decision Tree



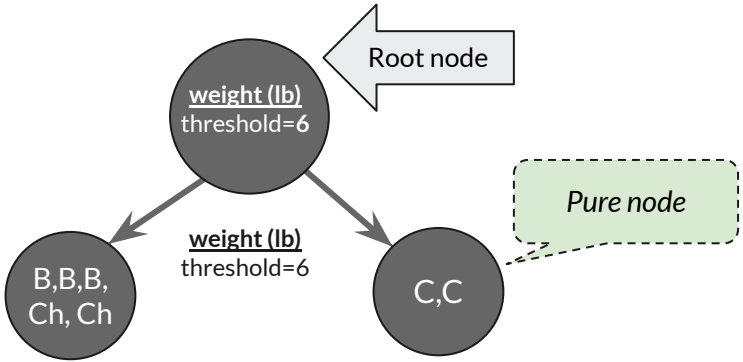
Test case

2	4	B

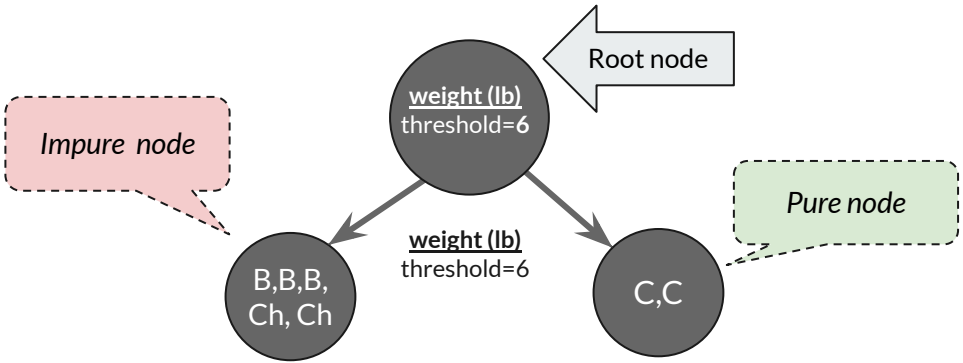
Decision Tree



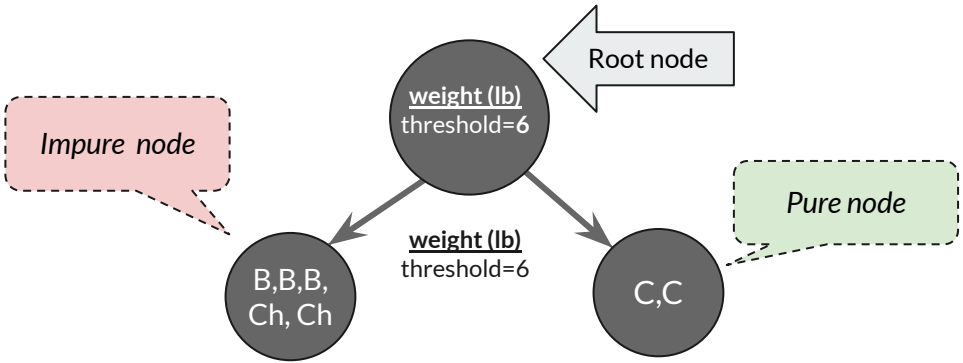
Decision Tree



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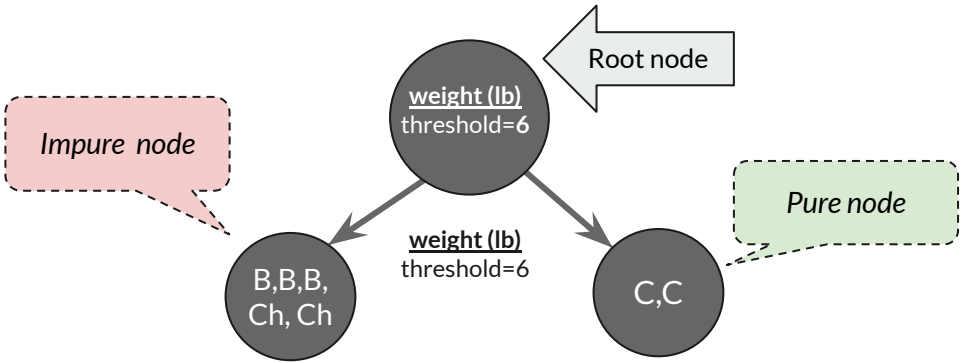
Decision Tree



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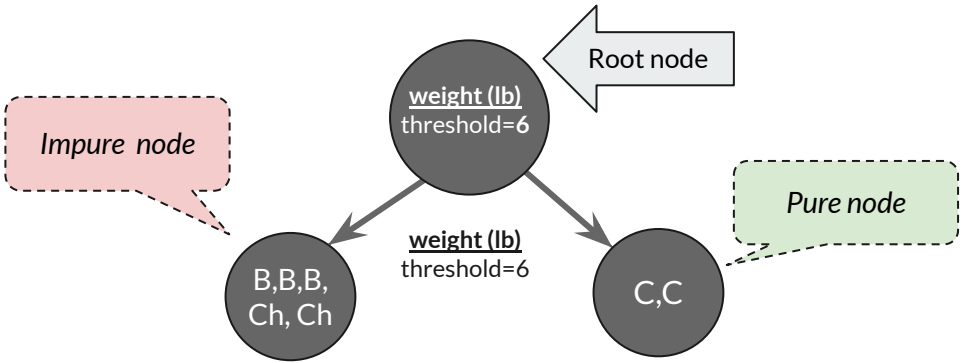
Decision Tree



Test case

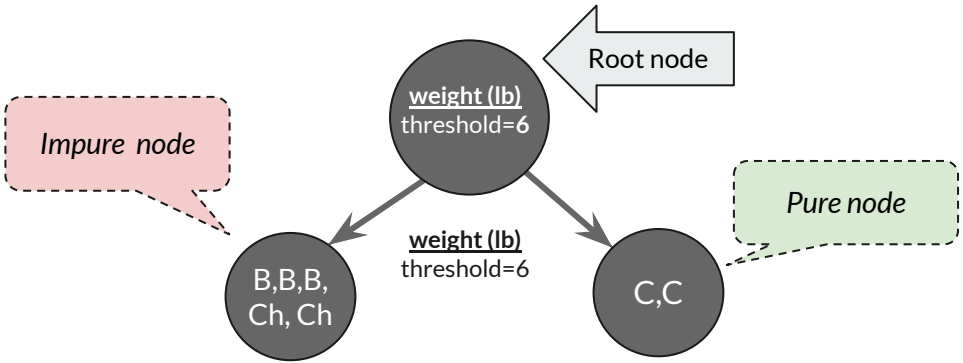
2	??	B

Decision Tree



Never can detect a ***Ch****(icken)*

Decision Tree



Never can detect a ***Ch****(icken)*

But has certain advantages (primarily in the context of ensemble learning, Bagging for an example).

Decision Tree -- Key Hyperparameters

Hyperparameter	Description
criterion	Measure of split quality (e.g., Gini, entropy, squared error).
max_depth	Maximum depth of the tree (controls overfitting).
min_samples_split	Minimum samples required to split an internal node.
min_samples_leaf	Minimum samples required at a leaf node.
max_features	Number of features considered when searching for best split.
max_leaf_nodes	Maximum number of terminal (leaf) nodes.
ccp_alpha	Cost-complexity pruning parameter (larger $\rightarrow$ more pruning).
splitter	Strategy used to choose splits (e.g., best, random).

Random Forest -- Key Hyperparameters

Hyperparameter	Description
n_estimators	Number of trees in the forest.
criterion	Split quality measure used in each tree.
max_depth	Maximum depth of each tree.
min_samples_split	Minimum samples required to split a node.
min_samples_leaf	Minimum samples required at a leaf node.
max_features	Number of features randomly selected per split.
bootstrap	Whether bootstrap sampling is used.
max_samples	Number of samples drawn per tree (if bootstrap=True).
n_jobs	Number of CPU cores used for parallel training.
oob_score	Use out-of-bag samples for generalization estimate.